Adrian Go Militante

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Professional Summary

Senior Test Automation Engineer with 15+ years of experience across medical devices, automotive embedded systems, semiconductor, and enterprise software. Known for building automation platforms from the ground up — from framework architecture and CI/CD pipelines through to team capability uplift. At WS Audiology, architected a C# and SpecFlow multi-layer framework covering hardware, desktop, web, database, and PDF layers; led an offshore team of engineers (initially 3, later 2); and introduced AI tooling including automated screenshot validation and an AI-scored interview assessment platform built on Azure OpenAI GPT-4.1 mini. At Continental, a Python-driven validation pipeline cut manual testing effort by 90%. ISTQB Certified.

Technical Skills

Automation & Programming:	Python, C#, C/C++, VB.NET, VBA, Selenium, SpecFlow (BDD/Gherkin), Appium Webdriver, Windows Application Driver, REST API, Postman, PdfPig, Flask
CI/CD & Version Control:	Azure DevOps (Classic & YAML), Jenkins/Groovy, Git, GitHub, Gerrit, Mercurial
Test Management:	JIRA, Confluence, HP ALM/HPQC, IBM Engineering Test Management, DOORS Next Generation, Azure DevOps Test Plans, Redmine
Embedded & Hardware:	CAN, I2C, SPI, Microwire, GPIB/VISA, Vector CANoe/CANalyzer/CANape/CANstress, Arduino, UTN, Loadbox
Test Infrastructure & Environments:	IIS, ASP.NET, VM Provisioning
Tools & IDEs:	Visual Studio, VS Code, LM Studio, Github Copilot
AI-Assisted Testing:	LM Studio / Qwen3-VL — vision-based UI validation, Ollama / Gemma3 — LLM-as-judge test evaluation, Azure OpenAI GPT-4.1 mini — AI-scored interview assessment

Professional Experience

Senior Test Automation Engineer

WS Audiology - WSA (Sivantos Pte Ltd) — Medical Devices | Singapore | Aug 2023 – Present

- Architected a unified multi-layer test automation platform from scratch in C# / SpecFlow BDD, enabling composable automated coverage across hardware, Windows desktop, web, database, and PDF layers.
- Integrated Arduino and UTN for remote hearing aid device control; automated the desktop workflow (configure → test via Appium Webdriver / Windows Application Driver) and result verification across two web applications via Selenium.
- Implemented SQL database validation to verify UI results against database state and PdfPig-based PDF content verification, all callable within the same SpecFlow step library.
- Eliminated code duplication across test runs and platforms via an INI file-based configuration system; developed reusable agent infrastructure — including a PC unlock mechanism — adopted across multiple projects.
- Built a Flask local report server to host test results; implemented automatic video recording attached to failing tests for faster root cause analysis.
- Established Azure DevOps pipelines from scratch in Classic Release, then upgraded to full YAML pipelines with expanded features; automated test plan and test suite management via Azure DevOps REST APIs.
- Identified a 3-day manual visual verification bottleneck; developed an AI-powered reporting tool using vision models (Qwen3-VL via LM Studio) and LLM-as-judge evaluation (Gemma3 via Ollama) to automatically validate UI screenshots, reducing turnaround from 3 days to 2 hours.
- Led infrastructure and framework setup for a dedicated API test project: provisioned VMs and physical machines mirroring production environments; built comprehensive MSTest coverage for 2 APIs with 8,000+ and 320,000+ test entries from CSV, both addressing prior production issues; configured detailed reporting for rapid identification of failing test cases; additionally set up infrastructure for the developers' MSTest system integration test environment.
- Championed Gherkin syntax standards across the team; conducted pull request code reviews to enforce quality baseline and mentor team members; created Postman performance test scripts for API validation.
- Built and led a team of automation engineers in India (initially 3, later 2): assessed capability gaps, defined coding standards, reviewed code, and guided root cause analysis on test failures.

- Built an AI-scored candidate technical interview assessment platform using Azure OpenAI GPT-4.1 mini, automating answer evaluation against structured rubrics and significantly reducing the manual effort of assessing candidates.

Senior System Test Engineer

Continental Automotive Singapore Pte Ltd (now Aumovio Singapore Pte Ltd) — Automotive Embedded Systems | Singapore | Apr 2018 – Aug 2023

Contract Apr 2018 – Apr 2020 · Permanent Apr 2020 – Aug 2023

- Designed and implemented a fully automated software validation pipeline in Python — covering installation, configuration (multiple options supported), test execution, and result delivery — triggered automatically by Jenkins CI on every new build.
- Automated software update testing across three methods: USB, Firmware Over-the-Air (FOTA) via Selenium-controlled web application, and CAN software loading via Python controlling the customer desktop tool — each producing automatic pass/fail reports.
- Integrated Vector tools (CANoe, CANalyzer, CANape) and a proprietary Loadbox simulating automotive switch controls, enabling hardware-software validation without manual bench operation.
- Automated stakeholder result delivery via email and integrated test results into IBM Engineering Test Management via Python, closing the loop between test execution and formal test records.
- Reduced manual validation effort by ~90% through full update testing automation; improved eCall test coverage by 30% in 6 months; cut DUT state measurement time from 1 day to under 1 hour.
- Developed defect reproduction scripts through root cause analysis, resolving 2 critical production issues in 2020.
- Maintained test server infrastructure (ASP.NET, IIS) and CAN simulations; authored Confluence documentation on automation processes and onboarded new engineers, reducing ramp-up time significantly.

Software Engineer

Gemalto Technologies, Inc. (now Thales Technologies Philippines Inc.) — Enterprise Software | Philippines | Jul 2014 – Jan 2018

- Developed a production automation tool that polls a shared network location for incoming files, processes them against internal company systems, and uploads results via Selenium.
- Built and maintained internal tools in Python, VB.NET, C++, and Selenium; performed UAT using HPQC, validating business workflows, documenting defect findings, and coordinating resolution with development teams.
- Coordinated the full software delivery lifecycle across global stakeholders: gathered requirements, managed defect resolution, provided end-user support, and led deployment events across international teams.

Test Development Engineer

Rohm LSI Design Philippines, Inc. — Semiconductor | Philippines | Feb 2010 – Jun 2014

- Created base test programs in C from the ground up for 3 new SPI and Microwire EEPROM device families, including test pattern development in assembly language, establishing the foundational suite for subsequent series device development.
- Automated a 15-item Japanese-standard code quality review checklist using VBA Excel macros, significantly reducing manual eyeballing and repetitive file operations while maintaining the engineering sign-off requirement.
- Developed parametric test programs in C/C++ for general-purpose and automotive EEPROM devices (I2C, Microwire, SPI); automated GPIB/VISA instrument control in VBA; provided line establishment and production support.

Education

B.S. Electronics and Communications Engineering — Honorable Mention

De La Salle University, Manila | May 2004 – Dec 2008

Certifications

- ISTQB Certified Tester – Foundation Level — Jul 2025
- Philippine Electronics Engineer (ECE) License — Oct 2009

References available upon request